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RESEARCH ARTICLE

Energy Efficient Path Planning Scheme for Unmanned Aerial Vehicle Using Hybrid Generic Algorithm-Based Q-Learning Optimization

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ABSTRACT Efficient path planning optimization strategies are required to maximize flying time while consuming the least energy. This research offers a novel approach for energy-efficient path planning for Unmanned Aerial Vehicles (UAVs) that combines a hybrid evolutionary algorithm and Q-learning while accounting for the UAV's velocity and distance from obstacles. To overcome the constraints of traditional optimization approaches, the hybrid methodology combines genetic algorithms and Q-learning. The suggested approach optimizes path-planning decisions based on real-time information by considering the UAV's velocity and distance from obstacles. Genetic Algorithm (GA) creates a wide collection of candidate pathways. In contrast, Q-learning uses reinforcement learning to make educated selections based on the UAV's present velocity and proximity to static obstacles. This integration allows the UAV to modify its path dynamically based on its energy requirements and environmental constraints. The main goal is to develop a UAV path planning scheme capable of dealing with obstacle-filled environments to improve energy efficiency and collision avoidance during flight missions. Our experimental results show that the hybrid technique outperforms the classical GA method in terms of energy efficiency by significantly reducing energy consumption while maintaining a suitable collision rate and the best path cost to the desired locations. The analysis results improve the performance of the hybrid GA/QL algorithm by more than 57.14% compared to classical GA.

INDEX TERMS AI, energy efficiency, path planning, unmanned aerial vehicles, energy optimization, hybrid genetic algorithm, Q-learning, obstacle avoidance, GA.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have received a lot of interest in recent years because of their diverse uses in domains like surveillance, monitoring, delivery, and aerial photography [1]. However, UAVs' limited onboard energy storage capacity continues to be a significant concern, limiting their operational capabilities and flying endurance. Path planning optimization methods must efficiently maximize flight time and minimize energy consumption, allowing UAVs to complete complex missions with limited resources [2]. Path planning algorithms for UAVs have

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traditionally depended on deterministic approaches, such as A* or Dijkstra's algorithm, which consider static environments and stated objectives. While these approaches can uncover plausible pathways, they frequently fail to optimize energy efficiency or adapt to changing mission requirements [3]. To solve these constraints, current research has looked into incorporating artificial intelligence (AI) approaches into path planning optimization, allowing UAV to make informed decisions based on real-time data and learning from previous experiences.

Swarm intelligence (SI) algorithms have emerged as powerful tools for optimizing the path planning of UAVs. Inspired by social insect collective behaviour and other natural occurrences, these algorithms provide decentralized and self-organized solutions to complicated optimization problems [4]. Generic algorithm (GA) and Particle Swarm Optimization (PSO) have been widely utilized to optimize UAV trajectories while considering issues such as obstacle avoidance, mission objectives, and energy efficiency. Ant Colony Optimization (ACO) replicates ant foraging behaviour and has been used to discover ideal UAV paths while taking topography, meteorological conditions, and mission requirements into account [5]. Another UAV path planning approach that considers dynamic environments and real-time limitations is Bee Colony Optimization (BCO), which is inspired by honeybee behaviour.

Hybrid swarm intelligence systems have shown potential in optimizing UAV path planning, particularly in complicated circumstances. Even with their achievements, issues like scalability and in-the-moment decision-making still arise. As a result, combining them with machine learning methods and attending to safety and security issues will improve path planning optimization and guarantee effective, flexible, and mission-focused UAV operations.

This study proposes a novel approach for energy-efficient path planning optimization in UAVs by integrating a hybrid genetic algorithm and Q-learning. The suggested algorithm uses both techniques' advantages to overcome conventional approaches' drawbacks. Genetic algorithms provide a global search by producing a wide variety of possible iterations. In contrast, using reinforcement learning, Q-learning makes selections depending on the UAV's velocity and proximity to obstacles. The program may dynamically adjust the UAV's course based on real-time input by combining these two strategies, maximizing energy consumption while considering mission-specific limits.

The primary goal of this project is to increase the energy efficiency of UAVs by developing an optimized path-planning algorithm that considers both global search capabilities and real-time decision-making. The hybrid genetic algorithm and Q-learning approach take the UAV's velocity and closeness to obstacles into account, allowing it to discover energy-efficient pathways that avoid collisions and optimize flying time. We intend to demonstrate our suggested strategy's higher performance and effectiveness through comprehensive simulations and comparisons with existing path-planning algorithms.

A. STUDY MOTIVATION

This study used a Genetic Algorithm (GA) and Q-learning combination to solve a UAV path planning energy efficiency challenge. These are some of the goals that this study can address:

- Minimize Energy Consumption: The suggested method enables the UAV to identify the optimum path that minimizes overall energy consumption, ensuring resource efficiency.
- Minimize Collision Rate: The Hybrid GA/QL technique allows for the calculation of the collision rate, which indicates the likelihood of the UAV colliding with objects, and the identification of pathways that minimize the collision rate, providing safe and collision-free navigation.
- Minimize Cost/Time: The Hybrid GA/QL assesses the fitness of everyone in the population, showing the cost or time required to achieve the goal, and allowing for the discovery of paths that minimize the cost or time required to complete the mission, optimizing efficiency and productivity.
- Optimize Path Planning: The use of GA and Q-learning together seeks to determine the best pathways for the drones. The goal may be to find paths that optimize several characteristics such as distance, velocity, and obstacle avoidance, resulting in efficient and successful navigation.

B. THE CONTRIBUTIONS

The study makes prospective contributions by conducting novel research using the proposed approach for energy-efficient path planning optimization.

- Hybrid Optimization Approach: Examine the effectiveness of combining GA with Q-learning in a hybrid optimization strategy for UAV path planning, and evaluate how the interaction of these two algorithms might lead to better solutions in terms of energy consumption, collision rate, best cost, and overall performance.
- The trade-off between Adaptive Exploration and Exploitation: Investigate methods for dynamically adjusting the Q-learning process' exploration and exploitation rates. Examine how an adaptive strategy can produce greater convergence and more effective path planning by balancing the exploration of new paths with the knowledge that has been acquired based on two parameters such as UAV velocity and its safe distance from the obstacles.
- Obstacle Avoidance: this allows for the incorporation of dynamic obstacle avoidance skills.

Overall, the main contribution is to advance energy-efficient path planning optimization for UAVs, with the potential to greatly improve operating capabilities, prolong flight

endurance, and enable effective resource utilization in a wide range of applications. We note that the proposed study is important because it addresses some of the challenges in the field of UAV technology, which are related to energy efficiency and the ability to adapt to complex environments. The proposed solution enables to improvement of path planning to reduce energy consumption by adjusting speed according to the safe distance between UAV and obstacles which may contribute significantly to building UAV systems and promoting responsible and sustainable use.

The rest of this paper is structured as follows: Section II presents a technical overview of UAV path planning optimization. The modelling and problem formulation are presented in Section III. Section IV examines the techniques and the proposed hybrid genetic algorithm-based Q-learning strategy. Section V provides the simulation setup and experimental results, and Section VI discusses them. Finally, Section VII provides the paper's conclusion.

II. TECHNICAL BACKGROUND

Much research has recently been done to address the issues of optimizing energy-efficient path planning for unmanned aerial vehicles (UAVs). Researchers have investigated various ways, including merging classical algorithms with modern optimization techniques and artificial intelligence (AI) methodologies [5]. Most combination approaches attempt to build algorithms capable of adapting to dynamic situations, avoiding obstacles, optimizing velocity control, and coordinating many UAVs [6]. These approaches seek to improve energy efficiency, mission capabilities, and the ability of UAVs to function effectively in various settings.

Recent studies have shown that swarm algorithms (SA) and machine learning (ML) algorithms can help UAV plan their paths more energy-efficient manner. These methods enable efficient computing while navigating complex settings. Additionally, these methods can be integrated to boost UAV path planning's energy efficiency and process optimization. Poudel et al., 2021 describe a hybrid path planning (HPP) algorithm for effective data collection by assuring the shortest collision-free path for UAV emergencies. The probabilistic roadmap (PRM) technique creates the shortest trajectory map in the proposed HPP scheme. The optimized artificial bee colony (ABC) approach is used to better distinct path restrictions in a three-dimensional environment. Regarding flight time, energy consumption, convergence time, and flight path, the findings of the results reveal that the proposed HPP beats the PRM and standard ABC schemes significantly.

Yuwen et al. 2021 provide the deep learning trained by GA (DL-GA) technique for addressing the issues of multi-UAV path planning in difficult settings. The authors emphasize the need for well-planned pathways to lower UAV numbers and energy consumption. The innovative DL-GA approach combines the GA and DL taking use of the speed and problem-solving skills of DL. GA takes data from many circumstances to train a deep neural network. This results in quick path optimization, especially suitable for scenarios with tight deadlines. The experimental findings show that DL-GA beats GA in solving speed without sacrificing optimization capacity.

Wang et al. 2022 consider the UAV's energy usage and the data gathering model. The proposed study aims to create a multi-objective UAV path planning model using improved particle swarm optimization and genetic algorithms. It suggests a hybrid method (IPSO-GA) to properly plan the UAV data-collecting path. Simulated trials therefore assess the effectiveness of this approach. The experimental results demonstrate that the hybrid method (IPSO-GA), based on enhanced particle swarm optimization and genetic algorithms can successfully build a path that effectively covers all data nodes while requiring the fewest number of UAVs.

Balasubramanian et al. 2022, the present study aims to develop an optimal energy-efficient path planning algorithm by combining the well-known Modified Ant Colony Optimization (MACO) algorithm and a Memory-Efficient A* Algorithm (MEA*) for obstacle avoidance in a three-dimensional (3D) environment and path access optimized for minimum power use. The proposed method combines the two effective algorithms mentioned above to improve the UAV path planning task. The hybrid method enables avoiding obstacles and finding an efficient path. During the simulation, the UAV attained the best trajectory during flight tests, and the total power consumption of the UAV was compared using the MACO-MEA* algorithm which obtained 21% lower power consumption and 55% faster execution time than the MACO-A* technique.

Dongcheng et al. 2022 proposed study focuses on quality-oriented UAV path planning and addresses local dynamic hierarchical planning and global static planning. Shorter planning timeframes and improved execution efficiency are achieved by developing a three-dimensional UAV path planning model and optimizing the A* algorithm, search strategy, step size, and cost function. Furthermore, the suggested scheme incorporates the dynamic exploration factor into QL to tackle the exploration-exploitation conundrum and facilitates local dynamic path modifications. When these methods are combined, an algorithm for global-local hybrid UAV path planning is produced. The suggested model and algorithm are useful for tackling path planning concerns associated with UAVs, such as dependability, security, and safety, as evidenced by simulation results that show how well they improve path quality.

Saeed et al., 2022 reviewed the features and characteristics of many swarm optimization techniques, including ant colony optimization (ACO), fruit fly optimization (FOA), artificial bee colony (ABC), and particle swarm optimization (PSO). Furthermore, the paper thoroughly assesses the most important papers on drone path planning algorithms. The authors concentrated on the swarm algorithm's influence and performance in drone path planning. They described one of the most commonly used algorithms to enhance the trajectory of drones relying on swarm intelligence, as well as its impact on the optimal path cost of drones. The results of the ACO

algorithm's performance study in a 3D and 2D-dimensional environment are presented and discussed, and then the ACO's performance is compared to the upgraded ACO algorithm. The suggested approach achieves fast convergence, hence speeding up the path-planning process.

Faten et al., 2023, give a thorough systematic analysis of the literature on path-planning methods for multiple UAS. The review focuses on biologically inspired algorithms in particular because of their potential to solve issues related to multiple UAV path planning problems. A taxonomy of existing algorithms is also offered, along with a description of their evolution in the literature. The study presents an organized and comprehensible overview of bio-inspired path-planning algorithms for academics interested in this field. The provided classification is significant because it allows for the study of different bio-inspired UAV path-planning algorithms within a single framework and highlights the main design elements of the algorithms to allow for extensive comparison.

Ahmed et al, 2023, provide an energy-efficient coverage path planning algorithm to handle the coverage path planning problem. The goal is to build a collision-free coverage path that consumes the least energy while covering the full region. During the simulation, the flight path was optimized and the number of rotations was lowered to save energy. The proposed method divides the ROI into cells depending on the UAV camera footprint. The coverage path planning problem is then developed, and the precise solution is determined. The proposed strategy provides a better solution in an acceptable period for small-scale problems. However, for large-scale issues, the technique fails to develop a solution in a reasonable time.

Alejandro et al., 2024, use Q-learning to self-adapt the drone tasks and improve path planning through obstacles-based map environment by learning from its successes and mistakes. The study used five different-sized maps, each with a unique set of obstacles and a different number of drones. The number of actions each drone took to complete the task in each trial was counted to determine the quality of the results. The suggested method yields findings indicating that the system achieves solutions with fewer movements as the number of UAVs rises. The results indicate that applying the Q-learning method in path planning suits working on maps with obstacles.

According to what was discussed in the related works, it is clear that the Swarm intelligence (SI) algorithms and ML used together to optimize energy-efficient UAV path planning have several benefits, including the ability to learn from data, tap into collective intelligence, manage complicated settings, and increase compute efficiency [6]. These methods can be combined to create effective and flexible path planning for UAVs, enhancing energy consumption and overall UAV performance. The various techniques shown in figure 1 can be utilized to enhance path planning for UAV.

In UAV path planning, various classical path planning methods have been widely employed. Most of these algorithms are concerned with determining the shortest or least-cost path from a starting point to a destination while considering static settings and stated objectives [7]. While they work well in simple circumstances, they frequently fail to adapt to changing impediments or optimize energy consumption. The following algorithms and methodology are employed in the UAV path planning strategy to optimize energy usage.

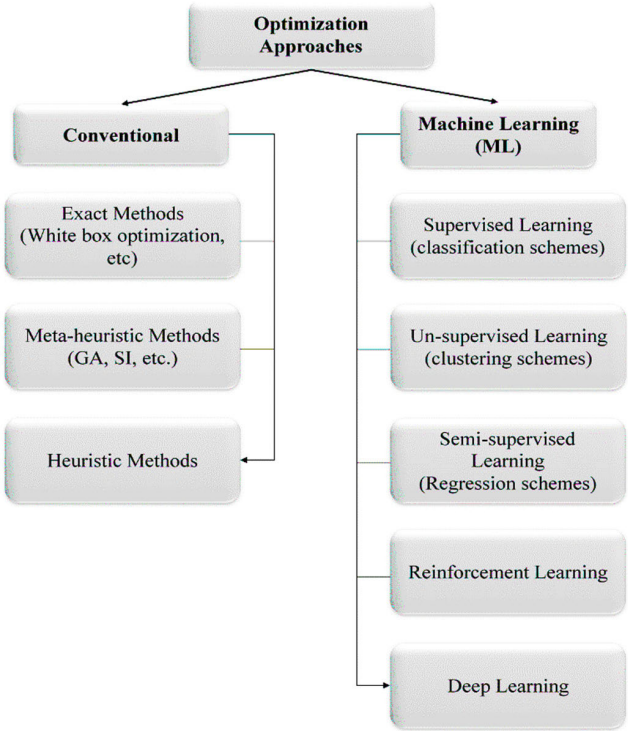


FIGURE 1. Intelligent approaches for energy-efficient path planning optimization.

A. GENETIC ALGORITHMS (GA)

Due to their capacity to operate in complex and dynamic situations, genetic algorithms (GAs) have become increasingly popular in UAV path planning optimization. For GAs to work, a population of candidate paths represented by chromosomes must be generated. These candidate paths must be repeatedly evolved using genetic operators like mutation and crossover [8]. GAs can search a large solution space and find energy-efficient paths by considering a variety of prospective paths and assessing their fitness considering predetermined objectives.

The GA uses genetic operators like mutation and crossover during evolution to produce new offspring by fusing and altering the fittest members of the preceding generation [9], [10]. Iteratively, the area covered and the energy used by each created path are considered during the fitness evaluation. The UAV energy consumption model considers flight direction, altitude, and speed variables. Up until a termination criterion, such as a maximum number of iterations or convergence of fitness values, is met, the GA continues to evolve the population, favouring individuals with higher fitness [10].

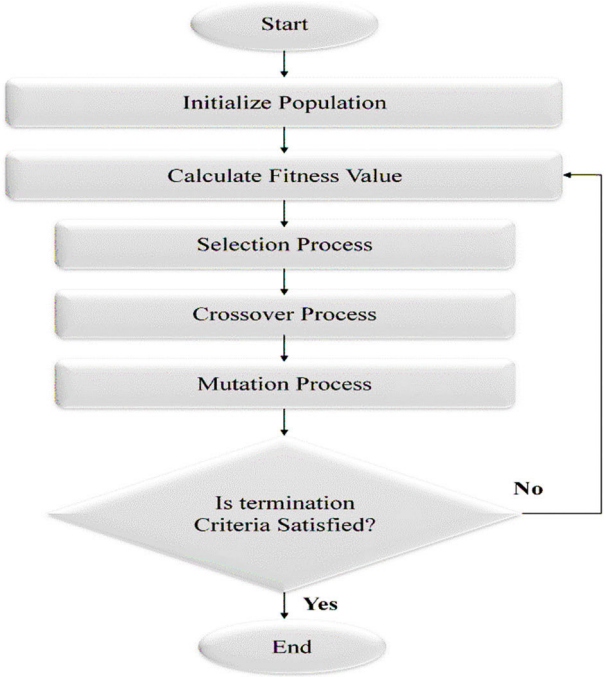


FIGURE 2. Flowchart of GA algorithm.

The fittest chromosome from the last generation represents the optimal course for the UAV to take to achieve an energy-efficient flight while covering the whole region of interest. The GA procedure is demonstrated in the steps below. The flowchart of a general algorithm is shown in Figure 2.

- Chromosome Representation: The GA chromosome is a series of waypoints the UAV must navigate. Each waypoint is denoted by W_i(x, y) or W_i(x, y, z) for the waypoint's 2D or 3D coordinates.
- Population Initialization: Create a population of chromosomes with a size of N and initialize it as P = [C_1, C_2,..., C_N], where each chromosome C_i denotes a potential path, the UAV could take.
- Fitness Evaluation: Assess the fitness of each chromosome in the population using the following fitness function.

$$\text{Fitness}\,(C_i) = W_1 \times \text{Coverage}\,(C_i) + W_2 \times \text{proximity}\,(C_i) \tag{1}$$

where, coverage (C_i) is the region of interest that is covered by following the path indicated by chromosome C_i, and proximity (C_i) is the distance between the path and any obstacles.

- Selection: Select parents from the population using a selection method such as tournament or roulette wheel selection with S selection rate.
- Crossover: To generate child chromosomes, perform crossover between selected parent chromosomes with a crossover rate of C.
- Mutation: To introduce random alterations, apply mutation to the offspring chromosomes, with a mutation rate of M. The mutation can include changing specific waypoints or adding/deleting waypoints.
- Fitness Evaluation (Offspring): Using the same fitness function as before, assess the fitness of the offspring chromosomes.
- Elitism: Keep the best individuals (elite chromosomes) from the previous generation intact in the next generation. An elitism rate of E determines the elite chromosomes [10].
- Termination Condition: When a termination criterion is satisfied, such as reaching a maximum number of iterations (Max Iterations) or achieving fitness value convergence [9], terminate the GA evolution process.
- Optimal Solution: In the final generation, the fittest chromosome reflects the optimal path the UAV should take to accomplish energy-efficient flight while covering the whole area of interest.

The UAV can effectively navigate the surroundings, avoiding obstacles and preserving energy throughout its mission using the GA for energy-efficient path planning. The ability of the GA to explore a varied collection of possible paths and optimize fitness based on both coverage and energy consumption allows the UAV to discover an optimal option that balances mission requirements with energy efficiency [10].

B. Q-LEARNING (QL) ALGORITHM

Q-learning, a type of reinforcement learning technique, has been used to optimize UAV path planning. Reinforcement learning enables the UAV to learn from its interactions with its surroundings, allowing it to make intelligent decisions based on rewards or penalties [11]. Q-learning algorithms learn the best policy that minimizes energy consumption and maximizes rewards depending on the UAV's current state by expressing the path planning problem as a Markov Decision Process (MDP).

The following stages are utilized in the procedure for the Q-learning algorithm in the context of UAV energy-efficient path planning [12]. The Q-learning flowchart is depicted in Figure 3.

- State Representation: The UAV's current condition during the path planning process is represented by the letter S, which stands for the UAV's status. The state may contain details about the UAV's location, speed, separation from obstacles, and other pertinent factors [13].
- Action Representation: Define the UAV's actions as A, which represents the range of potential actions the UAV may do in a specific condition. Actions that can be taken in path planning include choosing the next waypoint, modifying velocity, and altering altitude.
- Q-table: Keep track of a Q-table called Q (S, A) where the learnt Q-values for each state-action pair are kept. The Q-values show the anticipated future benefits of performing a particular action at a particular condition.

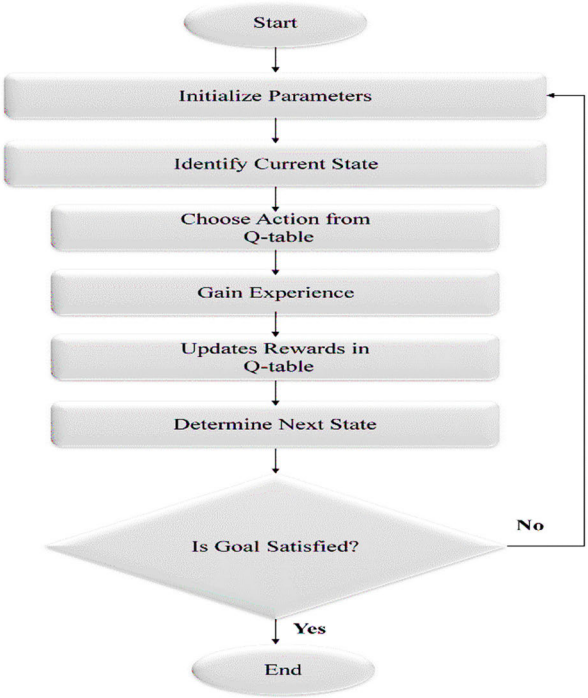


FIGURE 3. QL algorithm flowchart.

- Initialization: Initialize the Q-table with random values or zeros.
- Exploration vs. Exploitation: An exploration-exploitation trade-off parameter, such as epsilon (), is used to balance the exploration of novel behaviours and the utilization of learnt knowledge. Choose a random action with probability or, with probability 1-, the action with the highest Q-value during the learning process [14].
- Q-value Update Rule: Update the Q-values based on the Q-learning approach.
- Reward Assignment: Assign awards based on energy efficiency and task fulfilment goals. Assign good awards for efficiently reaching waypoints, and negative rewards for collisions or excessive energy use [14].
- State Transition: Based on the selected course of action and the existing environmental conditions, update the UAV's state. This entails modifying the position, speed, altitude, and other pertinent aspects of the UAV.
- Termination Condition: When a termination criterion is satisfied, such as reaching a maximum number of iterations or achieving Q-value convergence, the Q-learning process is terminated.
- Optimal Solution: The UAV is guided by the learnt Q-values in the Q-table to choose actions and make educated decisions that result in energy-efficient path planning and successful mission completion.

C. HYBRID GA/QL APPROACHES

To optimize UAV path planning, hybrid approaches that mix evolutionary algorithms and reinforcement learning methods have shown promise. These methods build various possible courses using genetic algorithms' global search capabilities and then apply reinforcement learning to adjust and refine the paths based on current data [15]. Hybrid algorithms may efficiently balance exploration and exploitation by combining these two methods, which enhances energy efficiency and adaptability.

The fitness evaluation is the foundation for a broad framework for executing the hybrid genetic algorithm and Q-learning approach in the UAV energy-efficient path planning context. The fitness value is the problem's optimal solution [16]. It uses the chromosome value with the best fit out of all those compared at each iteration. A new chromosome is formed for each cycle, along with its fitness rating.

Learning makes it possible to assess each chromosome's fitness within the GA population. The cumulative rewards earned throughout the Q-learning process can be used to assess fitness [17]. Update the Q-values based on the Q-learning equation using the hybrid technique afterward. It enables to expression of the state of the UAV using pertinent variables such as location, velocity, and distance from obstacles and balances the exploration of new actions with the exploitation of obtained knowledge using an exploration-exploitation trade-off parameter [18]. The Q-values based on the hybrid GA-Q-learning are updated according to the following equation.

$$Q\ (S, A) = (1 - \alpha) \times Q\,(S, A) + \alpha \times (R + \gamma \times \max\,[Q(S', A')]) \quad (2)$$

where, Q (S, A) represents the Q-value for state S and action A. α is the learning rate. R is the reward received after taking action A at state S. γ is the discount factor determining future rewards' importance. S′ represents the next state and A' represents the next action. max [Q(S′, A′)] represents the maximum Q-value for the next state.

D. REQUIRED SCHEME

Developing algorithms that solve obstacle avoidance with minimal energy consumption in real-time path planning approaches is a prominent priority in UAV path planning optimization. To enable UAVs to detect and avoid static obstacles during flight, the goal of the needed scheme must include a potential field-based approach, velocity obstacles, and reactive collision avoidance [19]. The proposed scheme considers the UAV's current position, velocity, and distance from obstacles to construct safe and energy-efficient pathways. The generated pathways are made up of straight-line segments that connect all the given control points a UAV must cover from start to finish [20]. Such a path is unsuitable for a UAV to follow since it necessitates immediate angular velocity and energy changes.

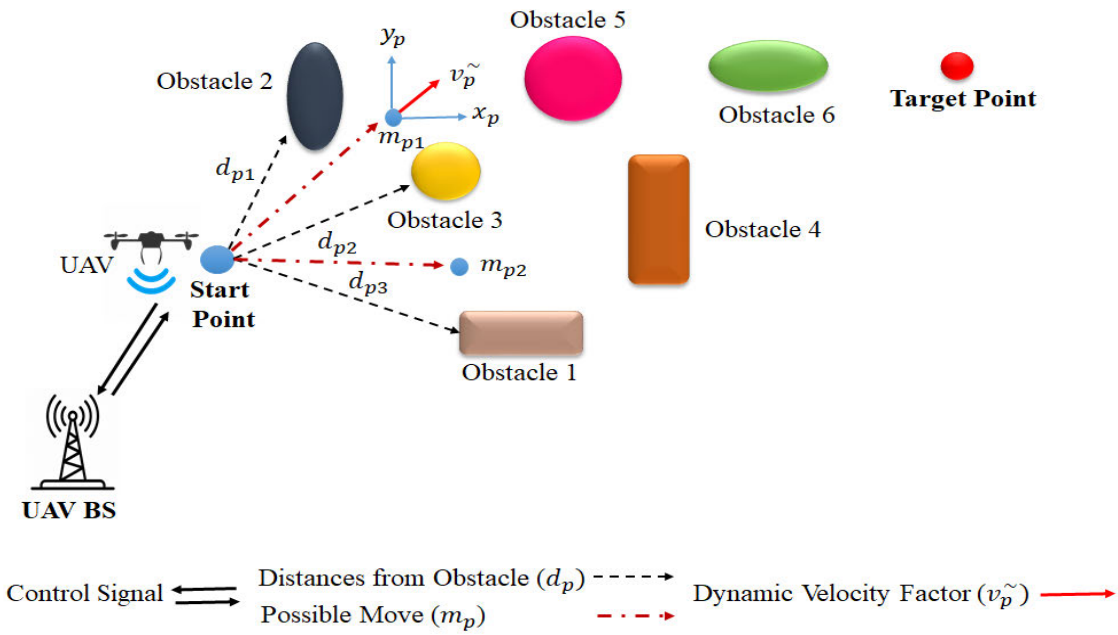


FIGURE 4. UAV path planning system model.

III. MODELING AND PROBLEM FORMULATION

The energy-efficient path planning problem in our suggested approach is established by specifying the waypoints the UAV will visit while travelling from start to end locations. The following sections describe our proposed study's general energy and optimized solution models.

The UAV path planning system model is shown in figure 4. The UAV is tasked with locating the target position in a 2-dimensional environment while avoiding collisions with objects and using as little energy as possible. The coordinate system (x, y) serves as the definition of the 2D environment space. Within this area, the UAV can move around freely. The UAV needs to reach a specified spot where the environment is known as the target location.

The UAV will move from its starting point toward its target while avoiding collisions with obstacles. The optimal path (shortest) for the UAV's travel is chosen depending on the distance from obstacles, and make velocity balancing at which it moves through them, which will achieve energy efficiency.

A. UAV GENERAL ENERGY MODELLING

By defining the energy consumption by ec_{ij} due UAV travel from point i to point j, the UAV travelling decision will take place according to i and j waypoints coordinates by the $\alpha_i\beta_i$ and $\alpha_j\beta_j$ respectively [20]. The following equation calculates the distance between the current UAV as i waypoint and possible future obstacles as j waypoint.

$$disntance_{ij} = \sqrt{(\alpha_i - \alpha_j)^2 + (\beta_i - \beta_j)^2} \tag{3}$$

The energy expended along this travelling path based on the distances between i and j waypoints denoted by x_{ij} [21] can be used to solve the energy-efficient path design problem.

$$\text{Min} \sum_{i=1}^{n} \sum_{j=1}^{n} ec_{ij} \times x_{ij} \tag{4}$$

where, e_{ij} be the energy consumption between waypoints i and j. and x_{ij} represents whether UAV at i will be assigned to the end location j. the ec_{ij} can be calculated by the following equation.

$$ec_{ij} = P_v \times t \tag{5}$$

where, P_v is the power required to perform travelling with velocity v across time t. In the suggested study, the UAV must calculate its velocity and distance from the obstacle to move in a safe environment. When the UAV travels along a path that consumes EC per unit distance, the total energy consumption cost can be estimated as follows [21].

$$EC_T = ec_{ij} \times disntance_{ij} + ec_{ij} \times t_v \tag{6}$$

where, t_v represents the time discretization when a UAV must hover for some time along its path to avoid colliding with obstacles.

B. ENERGY-BASED VELOCITY AND DISTANCE CONTROL MODELING

Our problem definition aims to reduce the overall UAV path planning energy consumption. The goal is to determine a path that gets to the desired location while balancing energy use, UAV speed, and closeness to obstacles. By setting additional variables of a given scenario, as t denotes the time index or discretized time step. v_t denote the UAV's velocity at time t. and d_t be the distance from the UAV to the nearest obstacle at time t.

The issue is determining how to define an energy model that ties the velocity and control inputs of the UAV to energy usage [22]. Let $E(v_t, u_t)$, where v_t is the UAV velocity and u_t is the control input such as power setting, representing the energy consumption of the UAV at time t. The UAV characteristics will determine this model, such as its power consumption.

A function that considers the UAV dynamics and control inputs might be employed to represent energy consumption. This issue is addressed by developing an objective function to lower the overall energy consumption on the track [23]. The total energy consumed throughout all time steps might be referred to as the objective function given by the following.

$$\text{Minimize} \sum_t E(v_t, u_t) \tag{7}$$

To change how important these variables are to the objective function, weights or scaling factors can be applied for velocity and the separation from obstacles. Let α and β be the scaling factors for velocity and distance, respectively. It is possible to express the modified objective function as follows.

$$\text{Minimize} \sum_t (\alpha \times E(v_t, u_t) + \beta \times f(d_t)) \tag{8}$$

where the function $f(d_t)$ measures how far obstacles are from the system and how much energy they consume. The exact relationship between distance and energy consumption determines the shape that $f(d_t)$ will take. The speed and distance constraints from obstacles are included to ensure UAV safety and feasibility during the route planning process. These constraints are as follows.

- UAV speed limits: $v_{min} \leq v_t \leq v_{max}$, where v_{min} and v_{max} are the minimum and maximum allowable speeds, respectively.
- Minimum distance from obstacles: $d_t \geq d_{min}$, where d_{min} is the pre-defined minimum safe distance.
- Other dynamics UAV limitations also will be considered including limitations on acceleration, communication cost, and Radius of UAV for collision detection.

According to the preceding equations, the goal is to reduce the number of rotations while minimizing energy consumption by optimizing the energy problem function based on the UAV velocity and distance from the obstacles [23]. To tackle our optimization problem, we employed a hybrid GA with QL technique to discover the optimum path with the least amount of energy. We assume in our suggested approach that the present UAV location to its end point calculations are straight lines in an obstacle-free environment. When computing collision-free pathways, the UAV is assumed to be in a 2D plane to consume the least energy and spend the least path cost [24].

IV. METHODOLOGY AND PROPOSED SOLUTION

An energy-efficient path planning technique is provided to develop solutions for optimal path planning with low energy usage. The algorithms utilized include GA and Q-learning, which allow the UAV path planning to be optimized to ensure a realistic collision-free path. We set the traveling coordinates for UAVs to (x_i, y_i) and (x_j, j) respectively. The suggested hybrid GA and Q-learning technique would allow the UAV to identify the best path-planning solutions while travelling through the provided coordinates. The best solution is based on optimizing the GA fitness function by Q-learning to learn and update the best solution results during population calculation and other fitness functions based on two criteria, the UAV velocity and distance from the obstacles. Algorithm 1 describes the proposed energy-efficient path planning computational approach.

The hybrid GA-QL algorithm begins with the population being initialized using the Genetic Algorithm (GA). Process in algorithm 2 evaluates the fitness of each chromosome in the population. The GA then evolves the population by performing selection, crossover, and mutation operations on the offspring chromosomes, resulting in fitter offspring chromosomes. The Q-table is then initialized using the fittest chromosomes from the GA [24].

The Q-learning method is used to select the best course of action for each state. The Q-value is updated using the Q-learning equation, considering the reward and the maximum Q-value for the following state. The UAV moves to the next state based on the action selected, and the process is repeated until convergence or the maximum number of iterations is reached. Throughout the process, the hybrid technique uses a -greedy policy to balance exploration and exploitation [25]. The state and action are updated, and the incentive is assigned based on mission fulfilment and energy efficiency. The stages in the flowchart are repeated until the termination condition is reached.

The proposed algorithm depicts the processes involved in the hybrid GA-Q-learning strategy for optimizing UAV energy-efficient path planning. It shows how the GA and Q-learning components collaborate to refine the solution iteratively and converge on an optimal path. Furthermore, when calculating velocity and distance fitness, the dynamic scaling factor technique is applied to improve the capacity to modify the relative relevance or weightage given to the velocity and distance fitness components. Algorithm 3 depicts the scaling factor approach technique.

The hybrid GA-based Q-learning optimization makes it possible to dynamically change the relative relevance of the velocity and distance fitness components based on the learned Q-values by applying this dynamic scaling factor to those components. The scaling factor rises as the Q-value rises, emphasizing the significance of the corresponding fitness component in total fitness assessment [26]. It is important to keep in mind that the precise formulation and parameters of the dynamic scaling factor may change based on the problem's demands and the Q-learning algorithm's unique characteristics. Experiment and adjustment may be required to determine the best values for k that produce the desired performance and balance between the fitness components [27].

Algorithm 1 Hybrid GA/Q-Learning Optimization

```
input: variables, UAV parameters, population size, max generations
set fitness, energy consumption, collision rate, and best cost arrays
set Q-table and UAV positions arrays
for generation=1: max generations do
        calculate travelling time;
        calculates GA fitness by Equation (1)
      set number of elite (elite percentage, population size)
        calculates the elite indices(fitness)
      find the next generation by population (elite indices)
      find the number of offspring (population size – number elite)
for i=1: number of offspring do
          use tournament selection to find parents & apply crossover to find a new solution
for i = 1: population size do
       calculates velocity, and distance
             state index = state index (velocity, distance)
       if state Index < exploration rate then
              action = get random action(population)
       else
              action = get best action (Q (state index))
       end if
          find new velocity & new distance based on action and new position
              calculating velocity and distance fitness by Algorithm 3
      if next state index <= number of states then
          find next Q value based on max Q value for next state index
      calculate reward = calculate reward (new distance) + calculate reward (new velocity)
         calculate Q (state index, action) by  Equation (2)
          set population (velocity) = new velocity; population (distance) = new distance;
      end if
         population (find (fitness == min(fitness), 1));
         Best offspring = offspring/next state index;
         Next Generation = concatenate (next Generation, best offspring);
                        evaluate fitness by Algorithm 2
      find    best population, best fitness
          calculate energy Consumption, collision Rate, best Cost, path Lengths
end for
end for
```

Algorithm 2 Evaluated Fitness Function

```
initialize function
input: communication Range, collision Radius, path Weight, communication Weight,
collision Weight
for generation=1: population do
      calculate energy consumption and collision rate
set    path cost = sum (population, 2);
calculate velocity fitness and distance fitness
   both fitness = velocity fitness + distance fitness
   evaluated fitness = both fitness + path weight × path cost + communication weight ×
   energy consumption + collision weight × collision Rate
end for
```

According to the demands or priorities of the challenge, the scaling of the fitness values emphasizes or de-emphasizes elements of the fitness evaluation process. The proportional weighting of velocity and distance fitness in assessing overall fitness can be customized using this scaling factor [28]. The GA evolves the population through its actions, producing

Algorithm 3 Dynamic Scaling Factor Approach

initialize state and action for Q-table using fittest chromosomes
for i = 1: population size **do**
If convergence is not reached and maximum iterations are not exceeded **then**
observe current state
choose action using Q-learning algorithm (based on ε-greedy policy)
update Q-value using Q-learning equation
transition to next state based on the chosen action
assign reward based on energy efficiency and mission completion
update state and action
end if
initialize Dynamic Scaling Factor Approach
Set scaling factor parameters k (magnitude), β(control), and Q(Q-value)
calculate dynamic scaling factor based on problem requirements or priorities
scaling = k × (1 – exp (-β × Q));
evaluate fitness = scaling × velocity fitness + (1 - scaling) × distance fitness
update velocity fitness using the scaling factor
update distance fitness using the scaling factor
end for

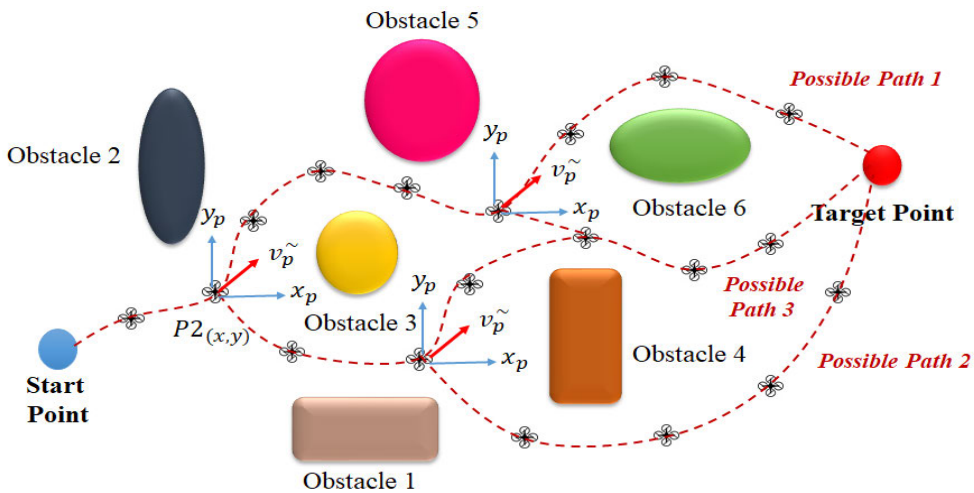


FIGURE 5. UAV path selection scenario.

offspring with fitter chromosomes and modified fitness values. In assessing the fitness of the offspring, the scaling factor influences how important velocity and distance are on a relative basis.

V. SIMULATION SCENARIO AND PARAMETERS

Given an obstacle-filled region of interest and a UAV, the goal is to discover an energy-efficient way for the UAV to navigate while covering the whole area of interest. The path planning problem is an optimization challenge to minimize the UAV's energy consumption while assuring mission success. We consider an area of interest in a 2D space, represented as a rectangular rectangle with dimensions (X × Y) for a specified scenario of UAV path planning optimization [28], [29], [30], [31]. Moreover, we assume 6 static obstacles in the area of interest, represented as polygons or circles with known positions and shapes. By setting that the UAV has a maximum velocity (V_{max}), maximum altitude (A_{max}), and a predefined battery capacity (B_c) that determines its energy limitations. Figure 5 shows the UAV best path selection scenario.

In the given scenario, the UAV must move and be able to navigate the region of interest using the best possible path. The best path (shortest) is a set of waypoints with (x, y) coordinates that completely cover the area and avoid hitting any obstacles. The scenario considers most constraints, those related to coverage, obstacle avoidance, and UAV operation, to account for the best path planning [32]. However, energy consumption should be considered as one of the limitations affecting the optimization of the path planning because the UAV can take a long time to travel if the route is not chosen considering the effect of energy consumption.

In the scenario, a hybrid algorithm is introduced to enable optimal path planning to avoid obstacles based on distance from the obstacles and to adjust the UAV velocity based

TABLE 1. Parameter simulation.

Parameters	Values
Simulation area	1000 ×1000 m
Number of UAV	1
Area dimensions	2D
Number of obstacles	6
Max safe flight speed	1.5 m/s
UAV altitude range	120 m
UAV mass	4.5 kg
UAV dimensions (length, width, height)	(1.8 x 0.3 x 0.2) m
Moments of inertia (x,y,z)	(0.1 x 0.1 x 0.045) kg·m²
Min safe distance from obstacles	6 m
Q-learning States	2 stats (velocity and distance)
Q-learning rate	0.2
Q-exploration rate	0.1
Velocity and distance scaling factor	Ranged between [0, 1]
GA, Hybrid GA population size	40
Selection probability	0.8
Crossover probability	0.7
Mutation probability	0.1
Energy calculations solutions	velocity and distance-based
Max iterations	1000

on the path and distance parameters. The UAV can identify potential collision risks by continuously measuring the distances from obstructions. The velocity is changed to steer clear of the obstruction when the distance drops below a predetermined threshold (min distance). The usage of hybrid Genetic Algorithm (GA)/Q-Learning, and distance-based velocity adjustment all help to discover the target's path efficiently [33], [34], [35]. The UAV tests out different paths, velocities, and techniques to arrive at the target in the shortest amount of time. The UAV can manoeuvre easily around obstacles, avoiding needless turns, and providing the shortest possible overall path through dynamic speed adjustments to ensure energy efficiency.

During its flight, the UAV must pass over the entire respective area. This can be described as a constraint that ensures that each point in the region of interest is a specific distance from the drone path. UAV must avoid collisions with barriers to comply with the obstacle avoidance limitation [36]. This can be described as a limitation that prevents the UAV's path from crossing any obstacles. Operational limits for UAVs' velocity, distance and capability must fall within predefined ranges. The simulation results presented in this study for evaluating the effectiveness of the proposed strategy are based on simulation scenario settings according to the criteria specified in Table 1.

VI. SIMULATION RESULTS AND DISCUSSION

The simulations make a comparison for the following evaluations.

- Evaluation experiment of optimization method GA and QL approach
- Evaluate the impact of Velocity and distance fitness dynamic scaling
- Compare the performance of hybrid GA-based Q-learning against GA for UAV optimizing the energy-efficient path planning approach

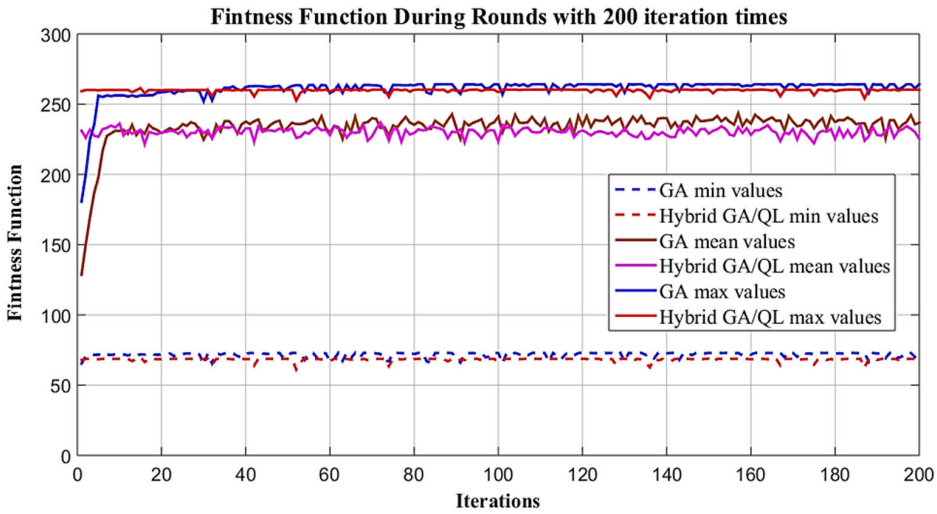
The testing and evaluation experiments are performed to assess the validity and effectiveness of proposed methods by using MATLAB 2020 as follows.

A. EVALUATION EXPERIMENT OF OPTIMIZATION METHOD

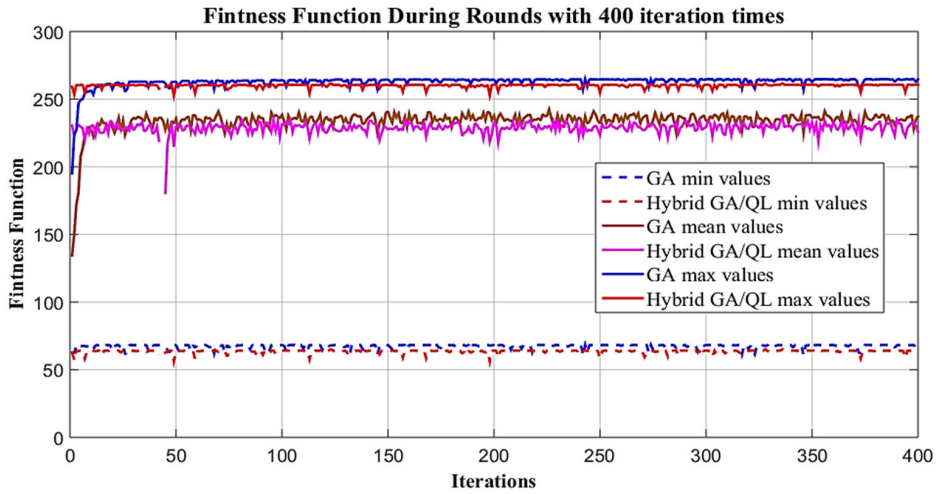
With the UAV task moving from start to end point, both GA and hybrid GA-based Q-learning optimization methods are simulated. The distance and velocity calculations are based on the chromosomal process with a population size of 40 at the start. The other GA parameters with the highest rate are as follows: mutation is set to 0.01, and crossover is set to 0.7. We obtained the path length and optimal fitness functions after simulating the algorithms with different iteration numbers of 200, 4002, 600, and 800, as shown in Table 2 [37]. The optimal fitness function represents the difference between the cost and revenue functions. In addition, the path length describes the path that the UAV takes in meters.

The results show that in all iteration numbers, the hybrid GA-based Q-learning provides a superior and optimal computation impact than GA. This suggests that the proposed hybrid GA-based Q-learning performs better computationally. Furthermore, the hybrid GA-based Q-learning continues to perform better as the number of iterations increases [38]. As can be seen, the fitness function for GA in 600 and 800 iterations yields almost the same value 0.4, but the hybrid GA-based Q-learning yields substantially better values ranging from 0.3 to 0.2. Both algorithms produce near values in all iteration numbers for path length. However, the hybrid GA-based Q-learning uses less path length, which means it performs better than standard GA [39].

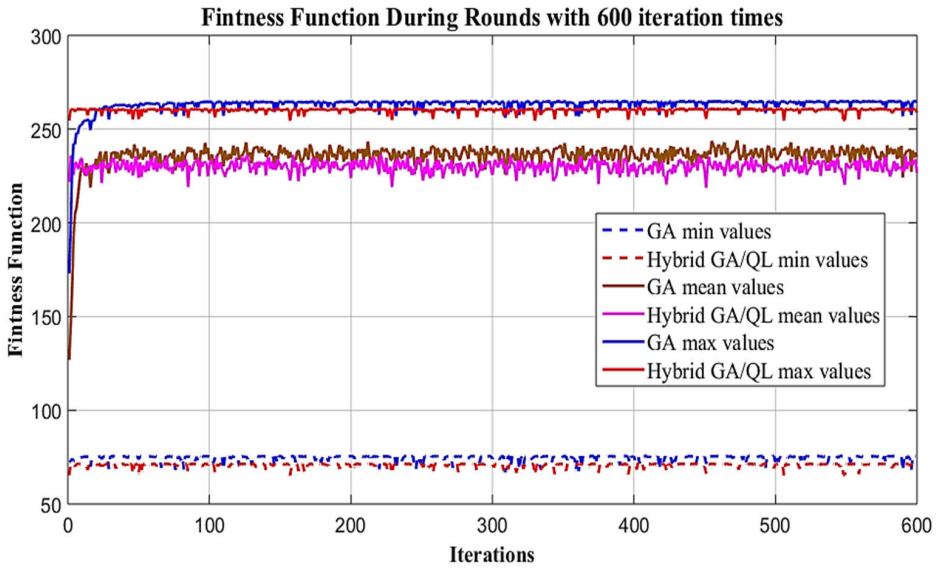
Furthermore, the results consider the computations of various fitness function average values (minimum, maximum, and mean) for both GA and hybrid GA-based Q-learning with iteration numbers of 200, 400, 600, and 800. Figure 6 shows how the suggested hybrid GA-based Q-learning improves performance during the UAV experiment.



(a) Statistical evaluation results of fitness function with 200 iteration times.

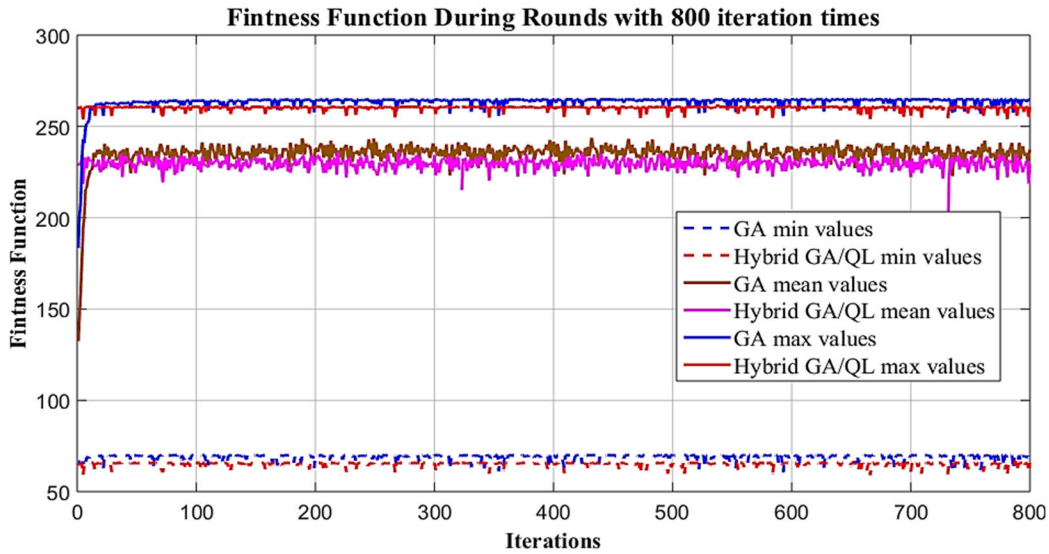


(b) Statistical evaluation results of fitness function with 400 iteration times.



(c) Statistical evaluation results of fitness function with 600 iteration times.

FIGURE 6. Average min, max and mean fitness function values of GA and GAQL using different iteration times.



(d) Statistical evaluation results of fitness function with 800 iteration times.

FIGURE 6. ***(Continued.)*** **Average min, max and mean fitness function values of GA and GAQL using different iteration times.**

TABLE 2. **Path length and fitness function for GA and hybrid GA-based Q-learning.**

Iteration Times	Scheme	Path length (m)	Optimal Fitness Function
200	GA without QL	197.0024	1.00
	GA with QL (Improved)	196.3736	0.9803
Iteration Times	Scheme	Path length (m)	Optimal Fitness Function
400	GA without QL	402.4296	0.6718
	GA with QL (Improved)	398.6169	0.5154
Iteration Times	Scheme	Path length (m)	Optimal Fitness Function
600	GA without QL	603.7753	0.4859
	GA with QL (Improved)	599.8708	0.3308
Iteration Times	Scheme	Path length (m)	Optimal Fitness Function
800	GA without QL	796.1990	0.4113
	GA with QL (Improved)	669.4261	0.2390

According to the statistical findings in (a), (b), (c), and (d) related to Figure 6, hybrid GA-based Q-learning allows the UAV to fly in a smooth flight path with high fitness function indices [40]. Figure 6 (a, b, c, and d) shows that hybrid GA-based Q-learning produces the best statistical fitness function values across all iterations. This indicates that our suggested method performs better when compared to traditional GA and allows most simulation times to locate the optimal options for path planning optimization.

B. IMPACT OF VELOCITY AND DISTANCE DYNAMIC FITNESS SCALING

The suggested scheme enables the hybrid GA-based Q-learning processes to dynamically modify the fitness scaling of UAV velocity and distance from obstacles according to the Q states [41]. The effects of the scaling parameters for velocity and distance from obstacles on the GA and hybrid GA-based Q-learning method for UAV energy-efficient path planning optimization are shown in Tables 3 and 4. When employing the dynamic fitness scaling with 1000 iterations, the evaluation of both GA and hybrid GA-based Q-learning is seen.

As demonstrated in table 3, dynamic changes in velocity scaling factors between [0, 1] with a constant distance factor of 0.5 positively impact the process of hybrid GA-based Q-learning, providing improved performance in total energy consumption, collision rate, and best cost. The same results were obtained in table 4 for the influence of a change

TABLE 3. Impact of velocity scaling factors with distance factor=0.5 in 1000 iterations.

Velocity scaling Factor	Scheme	Total Energy Consumption (J)	Collision Rate (%)	Best Cost
0	GA without QL	44.44	49.47	66.01
	GA with QL (Improved)	33.32	47.86	61.54
0.2	GA without QL	49.96	49.34	56.90
	GA with QL (Improved)	27.25	47.74	50.76
0.4	GA without QL	49.86	49.61	48.74
	GA with QL (Improved)	33.33	47.74	44.41
0.6	GA without QL	49.98	49.74	59.02
	GA with QL (Improved)	32.26	47.86	54.32
0.8	GA without QL	49.71	49.74	46.16
	GA with QL (Improved)	33.30	47.86	41.86
1	GA without QL	49.95	49.74	57.43
	GA with QL (Improved)	33.32	47.74	53.04

TABLE 4. Impact of distance from obstacles scaling factors with velocity facotr=0.5 in 1000 iterations.

Distance scaling Factor	Scheme	Total Energy Consumption (J)	Collision Rate (%)	Best Cost
0	GA without QL	49.89	49.74	53.92
	GA with QL (Improved)	33.32	47.74	49.55
0.2	GA without QL	33.32	47.74	49.55
	GA with QL (Improved)	33.32	47.86	58.54
0.4	GA without QL	49.81	49.74	63.56
	GA with QL (Improved)	33.33	47.86	59.24
0.6	GA without QL	49.94	49.74	58.15
	GA with QL (Improved)	33.27	47.74	45.98
0.8	GA without QL	49.95	49.74	63.43
	GA with QL (Improved)	33.29	47.86	59.05
1	GA without QL	27.17	47.63	49.60
	GA with QL (Improved)	33.32	47.86	47.99

in distance scaling factors between [0, 1] with a constant velocity factor of 0.5 [42], [43], [44], [45], [46]. The ability to change the relative priority or weightage assigned to these two fitness components in the overall fitness evaluation is improved by including a scaling factor in computing velocity fitness and distance fitness. The dynamic scaling of both velocity and distance scaling factors allows the UAV path to be evaluated based on the safe path and energy problem needs using valuable Q learning.

The results in Tables 3 and 4 show that modifying the proposed velocity and distance scaling factors will affect the balance between the importance of velocity and distance in the fitness evaluation [47]. By approximating the values of speed and distance while travelling from start to finish point over barriers, adjusting the measurement values for each of the two parameters determines the ideal values according to the path planning requirements.

C. PERFORMANCE OF GA-BASED QL PATH PLANNING OPTIMIZATION

In terms of total energy consumption, collision rate, best fitness, and best path cost, the overall performance of hybrid GA-based Q-learning for energy-efficient path planning optimization is obtained. Figure 7 compares the effectiveness of hybrid GA-based Q-learning to GA regarding overall energy consumption [48]. Because the Q-learning process can move the drone towards energy-efficient paths by accurately finding the learned policy, which depends on the quality of the reward function and the training process related to several factors such as the rate of UAV speed and distance to obstacles, the hybrid GA based Q-learning consumes less energy than GA [49].

Figure 7 shows that GA consumes more energy and is constant throughout iterations, whereas hybrid GA-based Q-learning consumes less energy, particularly at higher iterations. In general, the reduced energy consumption estimates obtained by Hybrid GA-based Q-Learning suggest their success in discovering more energy-efficient pathways for the UAV.

Figure 8 compares the collision rate results for hybrid GA-based Q-learning to GA. The collision rate results for GA range between 59% and 49%. These findings suggest that the GA method has a rather high collision rate during the path design process. The values vary, but they are often higher [50]. The Hybrid GA-QL technique, on the other hand, consistently shows a reduced collision rate, with values ranging from 48% to 47% over the same repetitions. These findings indicate that the Hybrid GA-based QL strategy is more effective in lowering collision rates during UAV path planning [51]. Lower collision rates imply that the pathways provided by the Hybrid GA-QL technique can better avoid obstacles and keep the UAV on a safer path.

Figure 9 depicts the best fitness for GA across 1000 iterations, with values ranging from 240 to 245 [52]. These findings suggest that GA can obtain somewhat acceptable fitness values during the path design. In contrast, the Hybrid GA-based QL strategy regularly outperforms GA, with fitness scores ranging from 237 to 241 over 1,000 repetitions. These findings suggest that the Hybrid GA-QL strategy is slightly more effective in identifying trajectories with higher fitness while optimizing UAV trajectory planning [53], [54].

Figure 10 depicts the cost findings for GA against the Hybrid GA-based QL method. Over 1000 iterations, GA shows a range of values from 46 to 70. These results show that GA achieves various cost values during the path planning process. However, the values fluctuate and in certain cases, larger costs are obtained [55]. On the other hand, the Hybrid GA-QL technique consistently displays lower cost values, ranging from 45 to 38 over the same 1000 cycles. These findings imply that the Hybrid GA-QL approach is more effective in finding lower-cost paths during UAV path planning optimizations.

Overall results show that in several areas of UAV path planning optimizations, the hybrid GA-QL strategy performs better than Genetic Algorithms (GA) [56]. In comparison to GA, hybrid GA-QL frequently delivers lower energy usage, collision rates, and cost values. This shows that the hybrid GA-QL method is superior at creating energy-efficient pathways that prevent collisions and maximize mission goals.

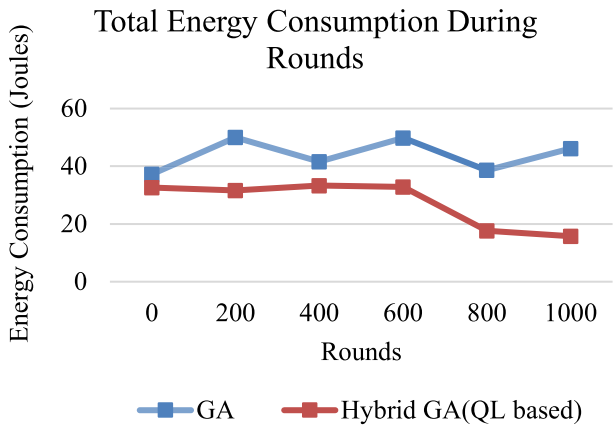


FIGURE 7. Performance of hybrid GA-based Q-learning in terms of energy consumption.

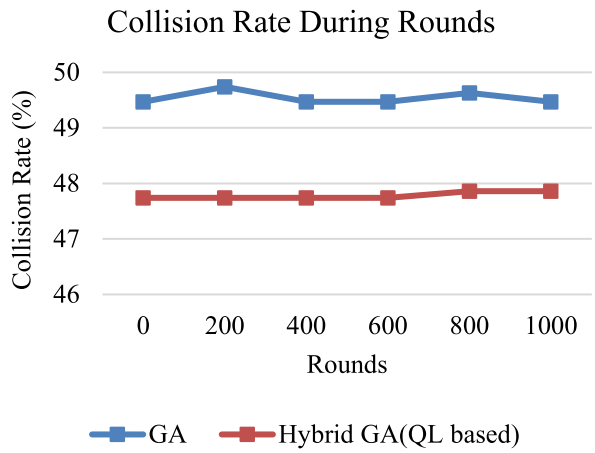


FIGURE 8. Performance of hybrid GA based Q-learning in term of collision rate.

D. PERFORMANCE LIMITATIONS AND POTENTIAL IMPACTS

According to performance results, the proposed hybrid GA-QL algorithm for UAV path planning shows good characteristics. However, it may also have several limitations and possible issues in real-world scenarios. First, in real-time applications, the computation and temporal complexity of the technique may cause decision delays, which may not be acceptable for time-sensitive UAV missions. The limitation lies in the requirement for specific tuning regarding the dynamic scaling factors according to UAV missions. Although the proposed algorithm shows that the dynamic factors adjusting will enhance the path planning, it is required to by intelligently optimized based on the real-time UAV mission.

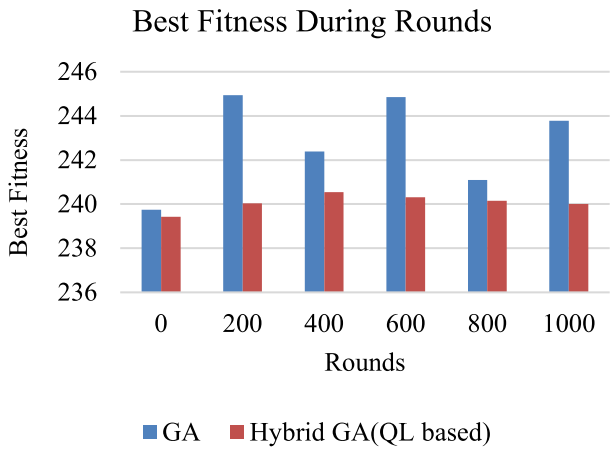


FIGURE 9. Performance of hybrid GA-based Q-learning in terms of best fitness.

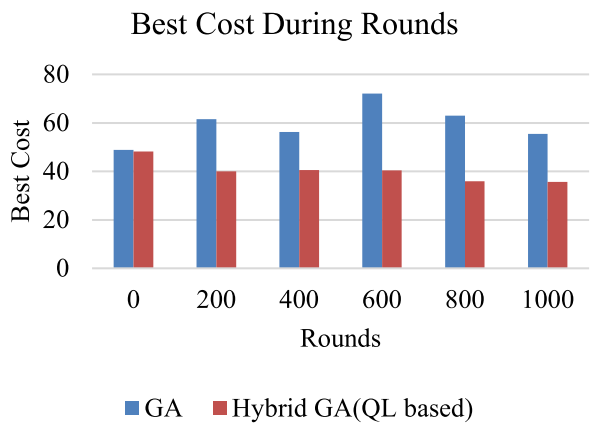


FIGURE 10. Performance of hybrid GA-based Q-learning in terms of best cost.

Secondly, one typical limitation of GA is the ability to adjust to unforeseen challenges or quickly changing surroundings may be constrained. Its responsiveness to real-time decision-making and real-world unpredictability may be hampered by QL reliance on pre-defined obstacles. The third issue is balancing energy efficiency with mission success or speed in addition to limitations related to scalability. We expect that the optimizing energy efficiency by the proposed algorithm can conflict with the requirement for speedy mission completion, even though it is essential for extending the UAV operational time. Balancing between meeting mission objectives on schedule and preserving energy must be carefully handled during real-time missions by adjusting the parameters related to GA, QL and dynamic scaling velocity and distance factors.

VII. CONCLUSION AND FUTURE WORKS

To plan UAV paths that are both energy-efficient and consider the speed and proximity of obstacles, this study introduces a novel hybrid genetic algorithm and Q-learning path planning optimizations approach. The suggested approach offers a practical response to the problems associated with UAV energy consumption while considering dynamic settings and mission-specific limits. The experimental outcomes highlight the hybrid approach's potential for real-world UAV applications in diverse settings by showcasing its energy efficiency and robustness superiority.

The innovative hybrid genetic algorithm and Q-learning path planning optimization approach for UAV path planning that was successfully developed displayed extraordinary capabilities in solving the dual issues of energy efficiency and collision avoidance. Compared to GA without optimization, our results demonstrate the outstanding performance of the hybrid technique in reducing energy consumption by 43% while retaining a low collision rate of 75%. Q-learning substantially contributes to the UAV path planning optimization scenario by enabling dynamic, energy-efficient, and safe navigation [57]. Its ability to learn and adapt based on previous experiences makes it an essential component of the hybrid genetic algorithm and Q-learning strategy, ultimately leading to improved results in terms of reduced energy consumption and decreased collision rates [58], [59].

Future research will concentrate on dynamic obstacle avoidance, optimal velocity control, integrating environmental parameters, multi-UAV coordination, and real-time adaptability for UAVs. By addressing these issues, progress can be made in developing algorithms that can handle dynamic obstacles, optimize velocity based on energy consumption models, adapt to changing environmental conditions, efficiently coordinate multiple UAVs, and dynamically adjust path planning in real-time. Deep reinforcement learning (DRL) can improve UAV path planning performance by addressing the limitations above by autonomously learning complex policies, making it well-suited for dynamic and unpredictable environments. DRL will help further improve energy efficiency and performance in UAV operations with the ability to address complex constraints.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request

CONFLICTS OF INTEREST

"The authors declare no conflict of interest."

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